PRIVOKE: A Client-Side Semantic Privacy Firewall for User Prompts in Web-Hosted LLMs

Nicholas Bisset*
Monash University Clayton, Australia
nbis0006@student.monash.edu

Manpreet Sahadevan, Nixon Ngai, and Sanoop Mallissery
Monash University Malaysia, Malaysia
{msah0018, nnga0004}@student.monash.edu, sanoop.mallissery@monash.edu

***Abstract*—Prompts sent to web-hosted language models can disclose personal and organizational information before a provider's controls apply. PRIVOKE inspects supported prompts on the user's workstation, decides whether to allow, warn or block, and enforces that decision before transmission. A separate service trains detector candidates from supplied labelled examples, checks version and retention constraints, and streams accepted parameters to clients. Independently randomized telemetry does not feed training. We evaluate detection and enforcement, retention during updates, and systems and privacy costs. Updated packages improve agreement on an authored contextual workload, but retain action failures and perform poorly on a separate annotation-presence task, where specificity is only 13.36–21.76%. All full-model candidates in a matched continuation study fail the retention guard despite aggregate gains. Installed-browser probes confirm bounded request handling; cost and telemetry audits expose measurement and estimator limits. The contribution is an implemented, auditable protection and update lifecycle. Independent contextual ground truth, practical utility and reliable generalization remain open.**

***Index Terms*—Prompt privacy, local inference, request enforcement, guarded model updates, local differential privacy.**

I. INTRODUCTION

Web-hosted large language models support writing, coding and analysis [1]–[3]. The context that makes a prompt useful can also disclose a person's health, a colleague's details or an organization's secrets. Reported workplace disclosures illustrate this risk [4], [5]. Once transmitted, sensitive content is already available to the provider; provider-side safeguards address a different point in the disclosure path [6]–[8]. A local intervention can inspect a supported request before it leaves the browser.

Local inspection alone does not resolve the problem. A detector can recognize an entity without knowing whether its disclosure is authorized. A correct warning may still transmit the original content. A detector update can improve average accuracy while losing a previously protected case. Even monitoring the system can disclose information unless its reporting path has a separate privacy boundary. These distinctions motivate PRIVOKE, a client-side prompt firewall that joins local inspection, explicit request enforcement, guarded detector updates and separately randomized telemetry.

PRIVOKE assigns contextual sensitivity, intended audience and information categories, then maps these findings to ALLOW, WARN or BLOCK. On supported requests, ALLOW and WARN forward the original body, while BLOCK cancels transmission. WARN attempts to display a notice without waiting for acknowledgement; it is not evidence of consent or prevented disclosure. The detector runs locally in the configuration studied here. A distinct server-side training path uses supplied labelled examples, validates candidate updates against their exact base, and distributes accepted parameters without rebuilding the extension. User prompts and telemetry do not supply training labels.

The closest work already establishes local prompt protection. Casper combines rules, entity recognition and topic detection in a browser extension [9], [10]; Hide and Seek trains local hiding and recovery models [11]; ProSan trains a local sanitizer that trades privacy hiding against task utility [12]. Our contribution is narrower than inventing local inspection or trainable protection: an implemented, version-bound detector lifecycle coupled to observable request actions, and evidence that distinguishes useful gains from regression and rejection. A retention check is a bounded gate, not a guarantee of safe adaptation.

We organize the evaluation around three questions:

1) **RQ1: Does PriVoke detect and enforce privacy-sensitive prompt policies effectively?**
2) **RQ2: Can PriVoke update its detector without sacrificing previously protected behavior?**
3) **RQ3: What systems and privacy overhead does PriVoke introduce?**

The answers are conditional. Authored contextual cases show substantial package gains and remaining enforcement harms; separate annotation-based tasks expose weak specificity. The update path works mechanically, but aggregate improvement does not ensure retention. Browser and telemetry measurements establish bounded behavior and costs rather than deployment utility. Keeping these answers together makes the system's contribution and its unresolved gaps explicit.

II. RELATED WORK AND CONTRIBUTION

Local protection and sanitization. Casper's three layers directly precede PriVoke's layered inspection [9]. Casper also substitutes identifiers, restores them in responses and offers topic-warning acknowledgement. It supports custom rules/topics and model replacement; it should not be characterized as uniformly static. Hide and Seek (HaS) anonymizes

*Corresponding author: Nicholas Bisset.
GitHub Repository Link

entities before cloud inference and locally recovers them from responses, with trained adversaries testing recovery resistance [11]. ProSan selectively replaces prompt words using task-importance and privacy proxies, then distils a local sanitizer [12]. These systems establish both local processing and training as prior art. PromptShield studies enterprise role/policy classification [13].

A different measured contract. PriVoke's browser interface acts on requests: BLOCK cancels a supported request, whereas WARN and ALLOW send its original body. Returned masked spans do not rewrite traffic, and there is no response-recovery stage. Its distinctive engineering focus is the guarded path from a candidate tied to an exact model revision to publication and client refresh. We measure paired gains and losses, including candidates that the guard refuses. The separate telemetry path limits randomized event fields without treating them as training feedback. These are bounded systems contributions, not claims of superior sanitization or uniquely local inference.

Comparison boundaries. The reviewed systems evaluate different targets. Casper reports entity/topic detection, HaS reports resistance to whole-text recovery plus downstream task proxies, and ProSan reports item replacement and automatic utility. Their published metrics cannot be placed on one privacy-accuracy axis. We retain version-qualified outcomes and interaction details in Appendix D. Our executed Presidio references use disclosed native entity-recognizer settings [14]; they are component references, not reproductions of Casper, HaS or ProSan. No matched system superiority claim follows.

III. System Design and Privacy Boundary

A. Scope and Trust

The aim is to reduce inadvertent personal, third-party or organizational disclosure through supported outgoing prompts. We trust the workstation, enabled extension, local runtime/supervisor and authenticated model publisher. Compromised clients, malicious publishers, deliberately replaced hooks and provider-held historical information lie outside this boundary. Recognized prompt POST formats through page `fetch` and asynchronous `XMLHttpRequest` are supported; workers, attachments, other transports and changed endpoints need separate validation. Disabling protection bypasses inspection. Optional hosted inference changes the disclosure boundary and is excluded from local-inference measurements.

Figure 1 separates three paths. Prompt inspection stays on the workstation. Labelled training examples enter a separate training service and accepted parameters return to the local detector. Coarse, locally randomized event fields go to a telemetry collector, with no path from telemetry to training. This separation describes what each service receives; it does not establish that every policy decision is correct.

B. Local Inspection and Request Enforcement

A page hook extracts supported prompt text and sends it through the extension background process and a loopback gRPC-Web bridge to a workstation-local Python runtime. A native-messaging supervisor manages bridge/runtime lifecycle. Local user inference does not fall back to the server-side training runtime. Figure 2 shows the request contract.

The runtime normalizes text while preserving offsets for original-text evidence. Its default cascade runs regular-expression rules first, followed by enabled named entity recognition (NER) and the semantic classifier. A rule-based BLOCK skips later detectors; other findings are aggregated with the strongest action retained. A failed requested layer blocks an otherwise ALLOW result, while valid non-ALLOW findings remain visible. Unavailable analysis blocks supported requests. A parameter-stream outage can leave other enabled layers operating, so fallback may reduce semantic coverage rather than block every full-pipeline request.

The semantic classifier returns three dimensions: sensitivity (none to highly sensitive), visibility (public to personal-private, or unspecified) and a set of ten information categories, such as health, identity and location. A deterministic policy converts these labels and confidence into an action. Highly sensitive content normally maps to BLOCK and potentially harmful personal information to WARN; identity/location combinations can raise restrictions. Low confidence can moderate the action. These labels and a confidence score do not independently establish disclosure authorization or calibrated risk. Appendix A gives the exact mapping.

We compare historical small classifiers with later packages retaining pretrained BERT encoders/poolers [15]–[18]. They classify prompts rather than generate conversational responses. Three package sizes are called efficient, balanced and quality; the names do not imply a universal performance ordering. Their tokenizers, capacities, thresholds and histories differ, so package comparisons do not isolate pretraining or architecture.

C. Guarded Detector Updates

The update process has two responsibilities: compute a candidate and decide whether that candidate may replace its base. A training controller constructs bounded labelled batches and held-out guard cases, excluding protected rows/text/groups. A separate server runtime fetches the specified model, executes training and returns model-bound parameter changes and candidate metrics. The updater checks exact base identity, allowed trainable tensors and shapes, finite values, magnitude limits and strategy-configured publication guards (Figure 3). A checksum or successful receipt establishes neither accuracy nor protection.

Validated artifacts are published atomically using compare-and-swap: publication succeeds only if the current base still matches the candidate's declared parent. Durable replay receipts make repeated submissions idempotent. Streaming serves versioned parameters; clients verify identities and refresh cached weights without an application release. This is centralized training from supplied examples, not federated learning, collection of user gradients or evidence of cloud deployment. The design enables updates; RQ2 asks whether particular updates preserve protection.

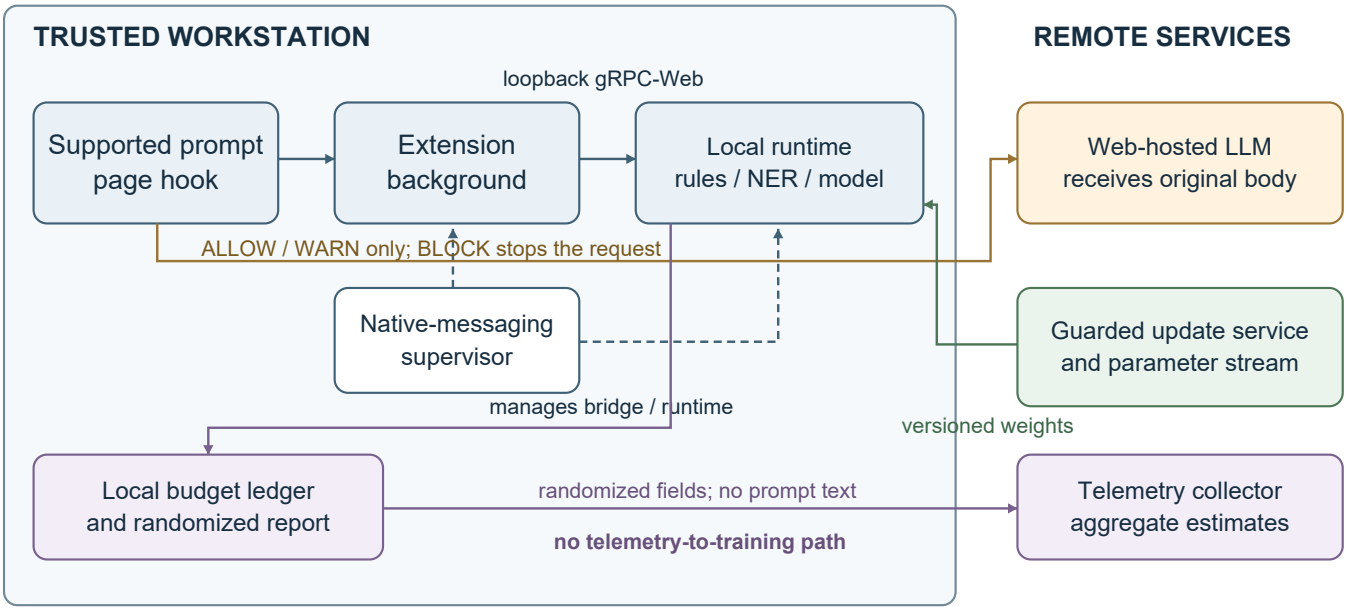


Fig. 1. PriVoke's three data paths. Supported prompt text is inspected inside the trusted workstation. Only an ALLOW or WARN decision releases the original body to the web-hosted LLM. Native messaging manages the local bridge/runtime; analysis uses loopback gRPC-Web. A separate update service supplies versioned weights, and a separate collector receives randomized event fields. Training consumes supplied labelled examples (Figure 3), not intercepted user prompts or telemetry. Dashed lines denote lifecycle control.

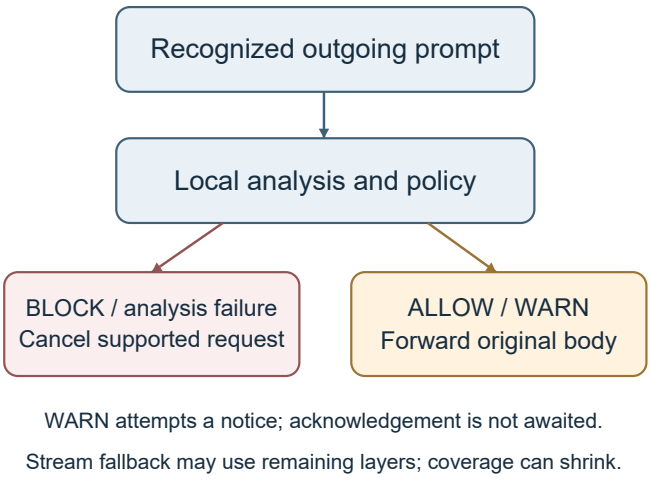


Fig. 2. Pre-transmission enforcement on supported hooks. WARN forwards unchanged content after attempting a notice; it does not wait for user acknowledgement. Analysis failure and reduced-layer stream fallback are distinct conditions. Disabled protection and unsupported traffic lie outside this decision path.

D. Separate Randomized Telemetry

Telemetry reports five coarse fields: action, an action/sensitivity risk bucket, one precedence-selected category, a bounded release label and a four-hour time-of-day bin without date. Packets exclude raw prompts, identifiers, exact confidence scores and per-layer findings. Each field is randomized locally before release. No telemetry field supplies a corrected label or feeds the training path.

For an event budget ε, each of five fields receives $\varepsilon_i = \varepsilon/5$. A field with K_i possible values reports truth with probability

$$p_i = \frac{e^{\varepsilon_i}}{e^{\varepsilon_i} + K_i - 1}, \quad q_i = \frac{1}{e^{\varepsilon_i} + K_i - 1} \tag{1}$$

for each alternative. Under ideal independent sampling and common public configuration, an event tuple has likelihood ratio at most e^{ε}. Actual finite arithmetic requires a separate audit. A persistent local ledger reserves budget before construction, defaults to event budget one and total eight per UTC day, and does not refund failures. Trusted clock/state are required. Report presence, contribution selection, delivery and network metadata are outside this content bound. The collector inverts nominal probabilities and clips marginal estimates; randomization can therefore impose substantial estimation error (RQ3).

IV. Evaluation Method

A. Questions and Evidence Units

Table I maps the existing experiments to the three questions. No experiment is counted as independent confirmation merely because it appears in more than one comparison. Detector execution is explicit: semantic-only calls request and verify only the semantic layer; product comparisons request rules, NER and semantics and retain legitimate rule-triggered skips.

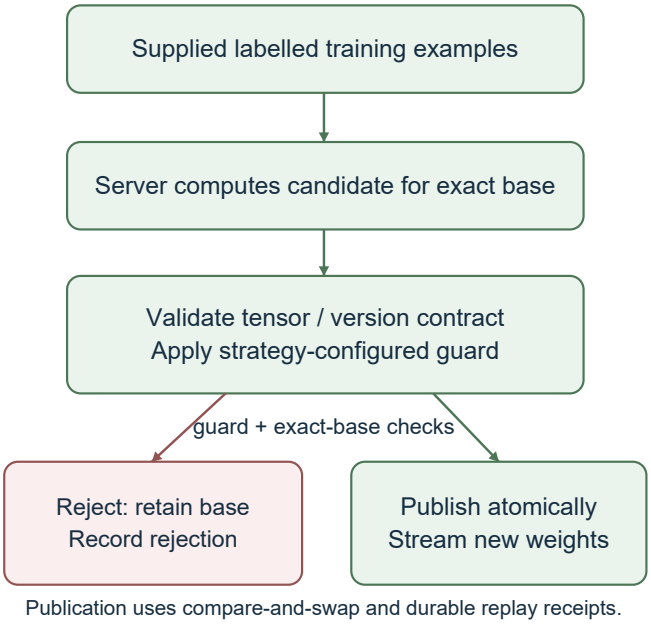


Fig. 3. A candidate is evaluated before publication. Compare-and-swap rejects stale/conflicting bases; durable receipts prevent an identical retry from applying twice. A passing retention guard covers its declared cases and criteria only. Head and underlying-model stages are separate publications, so a later rejection need not undo an earlier accepted head stage.

TABLE I
EVIDENCE UNITS AND THEIR LIMITS. PRESENCE/CATEGORY LABELS ARE AVAILABLE PROXIES; INDEPENDENTLY ADJUDICATED CONTEXTUAL PRIVACY TRUTH IS MISSING.

RQ	Evidence	What it establishes
1	1,560 authored contextual cases; 498 final presence rows; 2,000 external category rows	Agreement with distinct targets; not human privacy accuracy.
1	33 installed-browser receiver probes	Bounded request disposition; not provider coverage or usability.
2	Paired package outcomes; matched head-only/full-model continuation	Casewise gains/harms and guard decisions; not guaranteed adaptation.
3	360 browser decisions in 12 profiles; finite telemetry audit and synthetic simulation	Scoped systems cost and conditional privacy/estimation behavior.

Contextual policies. The authored workload contains 1,560 cases: 810 sensitive and 750 clean, with 260 per audience class. Sensitivity, audience, categories and allowed actions are assistant-provisional, without independent human adjudication. A binary positive means nonzero sensitivity or any category. Semantic joint correctness requires all three label dimensions; product joint correctness additionally requires an allowed action. An under-restricted action falls below the allowed-action set and an over-restricted action above it. These outcomes are distinct from receiver-observed transmission.

The workload has 35 scenario families and 24 audience-expression groups. It was sealed for an earlier semantic comparison, then opened and reused for product and recognizer studies. It is PriVoke-authored development evidence, not independent validation. Repeated passes establish repeatability, not a larger independent sample. Separate paired cluster resampling by family or expression describes dependence sensitivity; it does not certify population coverage.

Annotation-based checks. A once-only final endpoint contains 498 English rows with 236 annotation-positive and 262 annotation-negative targets from PIIMB revision `4a13e9f` [19]. Models, sources and scoring were frozen before release; no later tuning, reselection or promotion followed. This prompt-level presence task differs from contextual authorization and from PIIMB's official character-masking/F2 task. Its 464 observed source-group keys do not certify template independence. A separate external reference uses 2,000 AdvPI-IBench rows: 1,000 injected positives, 750 ordinary negatives and 250 hard negatives [20]. Official source categories are not exhaustive privacy annotations. Detailed selection, grouping and prediction proxies are retained in Appendix B.

B. Comparison and Retention Protocols

Historical and updated native packages are evaluated on the same contextual cases through the product handler, with unchanged rules and policy. Package differences remain confounded. A matched balanced-package ablation instead removes individual detectors while fixing texts, weights and policy; this isolates the local detector-removal contrast. Configured small/large English Presidio models provide same-text entity-recognizer references at threshold 0.5 [14], [21], with unsupported contextual/action outputs left unmeasured.

For RQ2, a fixed-base experiment compares head-only and full-model continuation across three seeds. Each treatment starts with byte-identical intact pretrained balanced encoder/pooler and zero task heads, receives the same ordered 4,096 training exposures over 64 steps, and is assessed on the same 1,080 reused provisional cases after complete training/guard group and text exclusions. Full-model continuation alone enables encoder optimization, including the consequent joint-gradient-clipping changes. All terminal candidates are assessed, including guard failures; no assessment selects a checkpoint. The 16-case retention guard requires joint improvement and zero action harms against its derived target actions. It is smaller and differently scored than the 1,080-case assessment, which permits declared allowed-action sets. Neither a weak zero-head control nor a small guard certifies broader safety. Recipes and historical unsuccessful branches remain in Appendices E and B.

C. Enforcement, Cost and Telemetry Protocols

Installed Linux Chromium probes use the built extension, standard native-host installer, native messaging, supervisor, bridge and historical balanced detector. An HTTPS synthetic receiver records original bodies and late sends. Thirty-three

TABLE II
SAME 1,560 PROVISIONAL CONTEXTUAL CASES, FULL PRODUCT. EACH ENTRY IS HISTORICAL/UPDATED. JOINT AND ACTION COUNTS HAVE DENOMINATOR 1,560; SENSITIVE RECALL COUNTS 810 AND CLEAN SPECIFICITY COUNTS 750. ZERO ERRORS IN EVERY ARM.

Package	Joint	Action	Sensitive	Clean
Efficient	266/693	1029/1415	544/808	494/629
Balanced	160/717	1026/1409	771/802	318/631
Quality	226/879	1073/1421	598/810	468/641

TABLE III
FINAL SEMANTIC ANNOTATION PRESENCE: 236 POSITIVE/262 NEGATIVE ROWS. HISTORICAL/UPDATED COUNTS; TRUE/FALSE POSITIVES (TP/FP), TRUE/FALSE NEGATIVES (TN/FN). ALL SIX ARMS HAVE ZERO ERRORS.

Package	TP	TN	FP	FN
Efficient	140/218	61/45	201/217	96/18
Balanced	141/221	65/35	197/227	95/15
Quality	167/190	51/57	211/205	69/46

correctness probes cover forwarding, blocking, faults, cancellation and recovery; this receiver is not an authenticated LLM provider. Full-pipeline and sole-semantic execution are checked outside resource windows.

The cost experiment requests rules/NER and measures 360 decisions across twelve fresh profiles, two per transport/action cell. Page-observed latency excludes receiver response. Whole-container CPU and sampled charged memory include resident infrastructure/cache and the observer. There is no unprotected baseline or incremental client attribution, and these windows do not measure semantic-layer cost. Appendix F specifies boundaries and sampling.

The telemetry audit enumerates finite sampling kernels for one fingerprinted host build, assuming independent uniform OS bits. Separate fixed-contributor simulations quantify estimator error, while real ledger checks exercise persistence, concurrency and failure accounting. These are conditional mechanism checks and synthetic utility measurements, not a deployed monitoring study.

V. RQ1: DETECTION AND ENFORCEMENT

A. Contextual Policy Agreement

Table II compares historical and updated product packages. All six runs completed without failed calls or layer errors. Updated joint correctness increased by 27.37–41.86 percentage points. Yet exact agreement remained 693–879 of 1,560, and gains did not preserve every previous success. Paired joint improvements/regressions were 484/57, 573/16 and 671/18 for efficient/balanced/quality. Allowed actions improved on 389/389/348 cases and regressed on 3/6/0. Quality's zero action regressions still concealed eighteen joint-label regressions. No package was promoted.

The cascade explains a recurring burden. All updated packages over-restricted the same 139 cases: 109 clean controls received rule-based BLOCK and skipped later detectors, while thirty low-risk controls received rule-based WARN. Rules ran on all cases and early-blocked 289; NER and semantics each ran on 1,271. Loaded NER produced zero converted PriVoke findings after supported-label filtering; this does not mean it found no raw entities or was inactive.

Removing rules from the balanced package increased allowed-action agreement from 1,409 to 1,530 and joint agreement from 717 to 972. It corrected all 139 shared unnecessary restrictions but lost eighteen previously correct actions; under-restricted cases rose from twelve to thirty. The 121 net action gains therefore coexist with lost protection under the provisional targets. Removing NER left full-product outputs unchanged on this fixture. Neither result justifies universally disabling a detector. All 10,920 ablation calls had valid execution records.

B. What Transfers Beyond the Authored Cases?

The separate final annotation-presence endpoint is a difficult counterpoint. All 2,988 semantic-only calls completed without errors and verified sole-layer execution. Table III retains the confusion counts. Updated recall was 80.51–93.64%, but specificity only 13.36–21.76%. Efficient and balanced specificity fell relative to their historical packages; quality's small increase was uncertain. Paired net-accuracy gain ranges from frozen source-group resampling were 7.41–17.37, 5.35–15.03 and 1.02–10.71 percentage points, respectively. Average gains therefore do not establish useful clean-label discrimination.

The same updated artifacts achieved 98.40–99.87% specificity on the authored semantic-only contextual fixture. The contrast holds artifacts and the nonzero-sensitivity-or-category response projection fixed, while changing texts and target labels. It is not an overnight model regression; its causes are unisolated. The mismatch between annotation presence and contextual labels limits interpretation, but does not remove the poor final result. Neither task supplies independently adjudicated contextual privacy truth.

The external source-category check shows a similar tradeoff under another proxy. Balanced updated recall was 92.9% versus historical 63.0%, but source-negative specificity was 0.5% versus 48.6%. This check counts any semantic finding as positive and labels only injected identifier categories, so 0.5% specificity cannot be restated as a proven 99.5% broad-privacy false-alarm rate. It remains poor agreement with the declared source target. Restricted five-type Presidio achieved 99.1% specificity with only 3.9% recall. All 6,000 model/recognizer analyses completed without errors; no threshold or row was changed afterwards.

On the authored contextual cases, small-English Presidio had recall 45.31% and specificity 71.87%, while its larger configuration traded recall down to 29.63% for specificity 88.80%. Both classified 906 of 1,560 correctly, with 192 paired gains and 192 losses between configurations. Updated product packages had recall 99.01–100.00% and specificity 83.87–85.47%, but their entity/contextual task mismatch and PriVoke-authored data preclude a general privacy ranking (Appendix C).

C. Does a Decision Control Transmission?

The complete installed-browser capture observed eighteen unchanged-body sends and fifteen zero-send outcomes across thirty-three declared correctness probes. Runtime-call correlation and late receiver checks supported the outcomes, without a decision broker. Healthy full and sole-semantic probes verified execution. An interrupted parameter stream blocked semantic-only requests; configured full-mode fallback executed rules/NER and forwarded, demonstrating reduced coverage. Restart restored full execution. Disabled protection/layers forwarded by design. Nine earlier failed captures remain separate, not pooled into the completed result.

Answer to RQ1. PriVoke detects many provisional sensitive cases and enforces its declared supported-request contract, including explicit blocking. Its effectiveness is conditional: updated packages still make action errors, WARN transmits original content, independent contextual ground truth is missing, and separate presence/category tasks reveal weak specificity. Receiver probes establish bounded enforcement rather than provider compatibility, prevented disclosure on every sensitive prompt or practical usability.

VI. RQ2: Updates and Previously Protected Behavior

A. Mechanism Versus Useful Adaptation

Audited server computation, bounded publication and client refresh demonstrate that served parameters can change. They do not establish that an update is worth retaining. Earlier contextual, curriculum and accelerated training studies failed their declared qualification criteria. A pretrained package improved provisional disclosure agreement but failed a serious-action veto. These negative results and their attempted budgets are retained in Appendix C, rather than displaced by later package gains.

Later supervised packages improved authored contextual classification, with component and action regressions retained. The RQ1 package comparison measures the resulting packages, not automatic online-training benefit. Its capacity, tokenizer, threshold and supervision differences prevent attribution to one update mechanism. The controlled continuation below asks a narrower question with a common base and exposure budget.

B. Matched Continuation and Guard Decisions

All six 64-step fits completed with finite parameters. Head-only training preserved every encoder tensor; full-model training changed 39 encoder tensors and recorded nonzero later encoder gradients. Seven snapshots, including the unchanged base, completed 7,560 native in-process semantic assessments without errors. These are repeated assessments of 1,080 situations, not 7,560 independent cases or deployed network calls.

Full-model continuation improved aggregate joint and action agreement relative to head-only training in all three seeds (Table IV). It nevertheless regressed on fourteen, twenty-six and twenty joint cases, and thirty-eight, forty-one and nineteen actions, respectively. Under-enforcement distance increased on 33/37/15 assessment cases and over-enforcement on 17/13/17; these counts are not a mutually exclusive harm union.

TABLE IV
Matched continuation on 1,080 reused provisional cases. Entries are head-only/full-model. Joint includes exact sensitivity, audience, categories and an allowed action. The unchanged zero-head base has zero joint and 480 action matches. Guard harms use sixteen separate cases.

Seed	Joint correct	Allowed action	Guard harms
42	192/257	908/993	0/1
43	204/247	906/989	1/1
44	198/268	912/1003	0/1

Every full-model candidate failed the fixed guard because of one action harm and would retain the unchanged base. Head-only seeds 42/44 passed, while seed 43 failed. These are hypothetical guard-to-base mappings: no candidate was published or promoted. The result illustrates why the publication decision must remain separate from the average score. The guard caught specific harms; its sixteen cases do not prove absence of other harms.

Answer to RQ2. PriVoke supports version-bound updates and can refuse candidates that violate a declared retention test. The evidence does not establish improvement without sacrificed protection. Matched full-model candidates improved averages yet all failed the guard, while package gains retained casewise regressions. Useful adaptation remains conditional on supervision, representative retention evidence and explicit acceptance criteria.

VII. RQ3: Systems and Privacy Overhead

A. Measured Systems Cost

The completed installed-browser protocol recorded 360 rules/NER decisions in twelve fresh-profile windows. Page-observed decision medians ranged from 27.9 to 35.0 ms across transport/action cells; each reported p99 is that cell's observed maximum. All 240 ALLOW/WARN decisions executed rules and NER, while 120 BLOCK decisions executed rules and legitimately skipped NER. These measurements exclude semantic-layer and receiver-response cost.

Across 126.40 seconds of summed windows, aggregate container CPU was 193.29 CPU-seconds, including 0.69 observer CPU-seconds. Maximum sampled charged memory was 1,267.91 MiB. Resident services/cache, driver and receiver share the container. A maximum sampling gap of 622.84 ms can miss memory spikes. Final-read and cleanup checks passed. Without an unprotected baseline or per-role attribution, these totals describe the instrumented system and do not isolate incremental PriVoke overhead. Startup intervals and cell tails appear in Appendix H.

Updated packages had lower observed host-call medians but larger server-process peak RSS than historical packages. Fixed execution order, warming, different backends and cascade shortcuts prevent a controlled speedup or model-only interpretation (Appendix G). Training times and allocated CUDA

tensor peaks likewise do not establish comparative training speed or whole-process memory requirements.

B. Privacy Cost and Telemetry Utility

The host finite-kernel audit found that maximum event privacy loss exceeds nominal budgets 0.5, 1 and 2 by approximately $(1.294, 1.503, 1.226) \times 10^{-15}$. Thus the exact nominal bound fails, despite the tiny excess. Eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same independent uniform-entropy assumptions. Other builds and deployed entropy remain uncertified. Appendix I distinguishes the implemented finite kernel from the ideal formula.

Randomization also reduces estimate usefulness. Fixed-contributor simulations found substantial clipping bias, root-mean-square error and incoherent marginal totals. For eleven-category reports with event budget one and eighty contributors, a true first-category count of seven had clipped bias +22.84 and RMSE 42.32 counts. Marginal estimates failed to sum to eighty in 99.99% of trials. This conditional synthetic result is not a population monitoring accuracy estimate.

Real ledger checks enforced the daily cap across persistence and concurrent requests, and failed construction/delivery still consumed reservations. Reversing a constructed event sequence changed first-eight BLOCK participation from 100% to 0%. Randomizing report contents therefore does not remove contribution-selection bias or protect whether an event participated.

Answer to RQ3. PriVoke introduces measurable local processing and infrastructure costs, but incremental and semantic-pipeline overhead remain unmeasured. Randomized telemetry has a conditional, narrowly exceeded nominal content bound and substantial synthetic estimation error. Neither telemetry's deployment utility nor protection of participation and network metadata is established.

VIII. Discussion and Limitations

A. What the System Evidence Supports

The central contribution is a complete protection lifecycle with explicit boundaries: inspect locally, enforce a request decision, evaluate a version-bound candidate before publication, refresh clients, and randomize separate telemetry. The experiments show why those boundaries matter. A semantic improvement cannot undo a rule-based early BLOCK. A WARN result does not prevent exposure. A larger average gain can be rejected for a single retained protection failure. Telemetry can support conditional aggregate estimation without becoming a source of training truth.

This contribution complements sanitization and recovery systems rather than replacing their utility questions. Casper, HaS and ProSan motivate what a user may need after detection: an acknowledged intervention, a usable rewritten prompt or a restored answer. PriVoke currently offers a request-action contract. It needs separate implementation and evidence before claiming those additional functions or superiority over them.

B. Ground Truth and Generalization Remain Open

The largest scientific gap is independently adjudicated contextual privacy truth. Our sensitivity, audience, category and action targets are assistant-provisional; they do not establish authorization or human agreement. Source annotation presence and injected category labels are useful checks, but cannot fill that gap. Entity detection, contextual disclosure, appropriate action and task utility need different labels. The large drop from authored clean specificity to final annotation-negative specificity is an observed limitation, not evidence that one target should be dismissed.

The authored cases are dependent and reused. Final source-group keys and external lexical exclusions do not certify template independence, cover paraphrases or remove unknown upstream pretraining exposure. Final rows are source-imbalanced: 308 Nemotron, 89 AI4privacy, eighty Gretel, twenty Privy and one MAPA. English synthetic/machine-generated workloads do not represent other languages or organizational policies. Separate cluster sensitivities are descriptive and unadjusted; repeated runs and three seeds do not create independent populations. Native-package differences further limit causal attribution. A new protected successor is needed for future tuning after the opened final endpoint.

C. Retention, Availability and Practical Utility

A retention guard can protect only what it tests. Our small matched guard and weak zero-head control leave considerable uncertainty. Rejected candidates are not served improvements, and the broader negative studies do not establish useful automatic adaptation. Future studies need common pinned bases, independent retention cases and explicit casewise harm criteria, retaining failed and effective artifacts separately.

Fail-closed behavior also has availability costs. Installed Linux probes cover declared historical-detector Chromium/native receiver paths; authenticated providers, Windows, updated-model installation, workers, attachments and other transports remain unverified. Confidence scores are heuristic, not calibrated joint correctness. WARN does not wait for informed acknowledgement. No study measures authorized task completion, unnecessary-intervention cost, warning comprehension or reformulation effort, so practical utility and human benefit remain unestablished.

Later instrumentation exposes ephemeral notice-rendering and local request-disposition events. Fifty-three synthetic host tests validate that contract, but the fields postdate the reported browser capture. They have no installed-browser or overhead remeasurement. Rendering does not prove comprehension, native handoff does not prove receipt, and page scripts can forge or suppress the events. They are not persisted or sent as telemetry; measuring utility still needs independent expected actions and joined user/task outcomes.

D. Measurement and Reproduction Boundaries

Cost scopes are deliberately narrow: instrumented container totals are not incremental client overhead, and rules/NER timings are not semantic timings. Telemetry bounds assume

retained local state, trusted clock and independent uniform OS bits. Delivery selection, participation and metadata are excluded, and synthetic estimator error has no deployment-usefulness threshold. Telemetry supplies no corrected contextual labels.

A sealed local source/model capsule reproduced twenty-four invented semantic requests across six packages with matching identities and outcomes. This is bounded replay using existing host dependencies, not public Git-only/fresh-environment reproduction or a protected-final rerun (Appendix I). Ignored pinned model/source assets and licensing constraints remain relevant to redistribution. These limits guide the next work: independent contextual adjudication, protected generalization studies, broader receiver coverage, baseline-relative semantic cost and human utility, rather than additional branches of uncontrolled fitting.

IX. Conclusion

PRIVOKE integrates local prompt inspection, pre-transmission request actions, guarded version-bound detector updates and separate randomized telemetry. The three research questions yield a qualified systems result: detection improves on authored cases but specificity transfers poorly; supported enforcement is observable, while WARN forwards original content; useful updates cannot be inferred from aggregate gains; and measured costs and privacy bounds retain substantial scope and utility limits. All full-model candidates in the matched study were rejected despite better averages. The system provides an auditable mechanism for protection and revision, without established independent contextual accuracy, safe automatic improvement or human utility.

APPENDIX A
EXACT POLICY REPRESENTATION

Sensitivity S0–S3 means no identified risk, low-risk personal context, potentially harmful personal information and highly sensitive/identifying details or secrets. Visibility P0–P4/PU means public, semi-public, authentication-restricted, group-private, personal-private and unspecified. The ten multilabel categories are HEALTH, POLITICS, RELIGION, CRIMINAL, FINANCIAL, SEXUAL, CHILD, LOCATION, IDENTITY and THIRD_PARTY.

The deterministic policy maps S3 to BLOCK and S2 to WARN. S0/S1 remain ALLOW unless IDENTITY and LOCATION coexist, or either occurs with P2–P4 visibility. Returned confidence below 0.5 moderates S3 to WARN and S2 without IDENTITY/LOCATION to ALLOW; other low-confidence classifications map to ALLOW. Aggregation preserves individual enforcement decisions, unions categories and takes the strongest sensitivity/known visibility. These are policy definitions, not independently validated disclosure authorization.

APPENDIX B
DETAILED COMPARISON PROTOCOLS AND TERMINOLOGY

Repository protocol identifiers below preserve traceability; main-text historical and updated packages correspond to V1 and AU3-selected V3. PH3 is frozen-encoder head fitting, EN3 joint continuation and AU3 audience-supervised rehearsal. PIPE3 is the full-product package comparison. HEADS/FULL denote head-only/full-model continuation. These names identify recipes and execution scopes rather than additional research questions.

A. Training Lineage and Selection

Historical online studies bind artifacts, curricula, trainable tensors, seeds, budgets and safety/retention predicates before assessment, retaining accepted and rejected attempts separately. Later paired semantic-only HEADS/FULL and representation studies use explicit execution scopes; publication does not imply qualification. Full specifications and unsuccessful attempts remain in repository research records.

PH3 fits task heads over frozen pretrained encoders; EN3 jointly continues encoder, pooler and heads; AU3 adds audience supervision with rehearsal. Validation alone selects checkpoints, without sealed-comparison tuning. Native V1 and V3 differ in capacity, tokenizer, threshold and history; selected task epochs exclude unknown upstream pretraining. Appendix E retains recipes, selected doses and parameter counts.

B. Matched Pretrained Continuation

A separate fixed-base host experiment compares HEADS and FULL continuation for balanced seeds 42/43/44. Each starts from byte-identical intact official pretrained encoder/pooler and zero task heads, distinct from both original V1 and AU3-selected packages. Complete TRAIN/guard group and normalized-text exclusions remove 480 AU3 validation rows before fitting, leaving 1,080 reused provisional cases (570 sensitive/510 clean) in 25 families. Each treatment receives the same ordered 4,096 TRAIN exposures over 64 steps, sensitivity/audience/category objective and dropout-off execution. AdamW head/encoder rates are $10^{-3}/2 \times 10^{-5}$, weight decay zero, clip norm one and microbatch 4/effective batch 64; FULL alone enables encoder gradients and its optimizer group, including consequent joint-gradient-clipping differences. The two-step pilot establishes execution/resource readiness only.

All six terminal candidates and one unchanged ORIGINAL are assessed through native in-process handler requests with explicit successful sole [4] and immutable identities. In this continuation study, joint correctness also requires an allowed action in addition to exact sensitivity, audience and category set; executed layers remain semantic-only. These are 1,080 cases assessed seven times, not 7,560 independent situations or network RPC evidence. No assessment selects a checkpoint. The maintained 16-case guard requires joint improvement and zero action harms against one derived target action; assessment instead permits its declared allowed-action sets on 1,080 different cases. Guard candidate/ORIGINAL mappings are hypothetical, with no publication. A bounded guard pass does not certify broader policy safety. Three seeds share one weak zero-head control and the same cases.

C. Product and External-Reference Protocol

PIPE3 compares original V1 and AU3-selected V3 efficient/balanced/quality packages through the native regex-first handler and a narrow inference-only V3 adapter. Requests explicitly select [2,3,4], no audience hints and no presence gate; identities, actual layer status and errors are checked. Real installed regex/spaCy detectors are used. First-pass same-case accuracy is primary. Three passes/package yield 28,080 accuracy RPCs plus 24 separate warmups; later identical passes measure repeatability and do not increase independent sample size.

A separate product-scoped ablation fixes the AU3 balanced package, rules, NER, policy and all 1,560 reused cases. Seven explicit configurations request regex [2], NER [3], semantic [4], nonsemantic [2,3], no NER [2,4], no regex [3,4] and full [2,3,4]. Actual execution, legitimate regex skips, identities and errors are checked for every request. This is a matched detector-removal comparison for one package, separate from semantic-only model evaluation. Native V1 windows 64/96/128 and category threshold 0.38 are retained; V3 uses 512 WordPieces and threshold 0.5. Three rows exceed V1 efficient's cap: one regex-skips semantic and two use native truncation; none is removed. Windows, tokenizers, capacities, thresholds and histories prevent causal architecture/pretraining attribution. The CPU host uses Python 3.11.9, Torch 2.11.0, NumPy 1.26.4, spaCy 3.7.5 and `en_core_web_sm` 3.7.1 with four model threads. V3 runs before V1 in fixed tier order, one loaded process at a time. Host RPC includes product/transport overhead; peak RSS covers the server process.

Fixed order, backends, warming and regex shortcuts make cost descriptive.

Independent Presidio 2.2.363 uses the same ordered texts, spaCy 3.7.5, native recognizers and threshold 0.5 [14]. Separate fixed configurations use `en_core_web_sm` 3.7.1 and `en_core_web_lg` 3.7.1 [21]; the larger-model follow-up was frozen before its predictions. Any finding predicts positive. Only text/language enter inference: no target hints, PriVoke rules, mapping, tuning or exclusions. Its entity task deliberately differs from contextual binary truth. Severity, audience, category exactness, joint and actions remain unmeasured. Raw same-ID pairing uses existing PIPE3 first passes without rerunning them. The larger-reference freeze omitted an imported analysis-helper hash; independent raw confusion/pairing checks address the reported counts without retrospectively completing that freeze. Direct analyzer timing is not compared with product RPC latency.

Source closures, package/data/recognizer hashes, raw journals and independent audits bind the studies. The tracked compact aggregate/generator reproduce figures without prompts or weights; inference reproduction additionally requires ignored pinned assets, including historical dirty source bytes. A post-hoc raw audit reconciled all 154 declared source files and eighteen model assets to their pinned historical bytes. Relative to clean commit `65a7c78`, 73 source files match exactly, 55 differ only in line endings, eleven differ beyond line endings and fifteen are missing; a byte-exact local archive preserves the required closure. Binary recall, specificity and accuracy use 2,000 paired whole-cluster draws, seed 42 and linear 95% percentile ranges separately by family or audience expression; these describe dependence sensitivity rather than crossed-cluster independence or population coverage. Ten fixed equal-width bins compare returned semantic confidence with observed sensitivity/category exactness where exactly one scored semantic finding was emitted. This outcome-selected subset excludes regex skips and semantic executions without an emitted finding. The scalar takes the maximum sensitivity/category-head probability; it is neither presence nor full-correctness probability, so ECE/Brier calibration metrics are not inferred.

D. External Official-Category Reference

A prospective AdvPIIBench reference [20] freezes revision `02741d9f` and 1,000 injected positives, 750 ordinary negatives and 250 hard negatives. The source is English machine-generated single-turn data without human annotation; only the attacked identifier is labelled, so it does not establish exhaustive five-type or broad privacy-clean negatives. The target is agreement with official CATEGORY, distinct from contextual privacy. Preparation builds whole-source closure over parent IDs, exact/normalized text and typed original injected values before exclusions, then selects globally one row per component, excluding few-shot-safe and protected/conflicting components. All quotas and protected TRAIN/validation/bootstrap exclusions passed before inference; the 2,000 selected components are not certified independent templates.

For this reference alone, any returned semantic finding predicts positive, including WARN; this response proxy differs from the non-S0-or-category projection above. Balanced V1/V3 use explicit successful sole [4] and unchanged native settings. V1 truncates 477 inputs at its 96-token bound; V3 has no over-cap input at 512, precluding training-only attribution. Presidio lg is separately restricted to credit card, phone, IBAN, email and US SSN at threshold 0.0, unlike the earlier all-recognizer threshold 0.5 reference. Paired component resampling uses 2,000 draws, seed 20261014; one row per component makes it row resampling here. Neither finding presence nor CATEGORY agreement measures attacked-span recovery or human privacy accuracy.

APPENDIX C
RETAINED UPDATE AND REFERENCE DETAIL

The unsuccessful branches reported under RQ2 are separate protocols, not a pooled success rate. Early contextual grids completed 126 attempts without a retained qualifier; the curriculum/representation study completed 900 live attempts and 360 offline steps without qualification. None of nine revised realizations in the accelerated 2,016-attempt study met its promising criterion. Disclosure joint correctness remained zero throughout the accelerated 2,016-attempt experiment. The pretrained MiniLM package improved provisional disclosure agreement but failed its serious-action veto. On the sealed AU3 semantic comparison, EN3-to-AU3 joint classification rose from 262/267/429 to 922/972/1209 of 1,560 for efficient/balanced/quality. Paired improvements/regressions were 680/20, 732/27 and 792/12. Clean matches were 738/740/749 of 750. Semantic joint here excludes action; product joint includes it. Quality retained classification regressions despite no action distance; efficient/balanced acquired new under-restriction harms. These supervised package outcomes do not establish automatic online-training benefit.

Small-English contextual Presidio confusion was TP/TN/FP/FN 367/539/211/443; large-English was 240/666/84/570, both using native recognizers at threshold 0.5. Updated product efficient/balanced/quality paired binary improvements/regressions versus the small reference were 612/81, 607/80 and 613/68. Historical 502-row development annotation presence instead yielded original-product recall/specificity 93.56/19.75% and configured Presidio 78.79/79.41%; these are different targets and snapshots.

External source-category balanced V1/V3 TP was 630/929 of 1,000; TN 486/5 of 1,000, divided into ordinary TN 321/5 of 750 and hard TN 165/0 of 250. Five-type Presidio returned TP 39, TN 991 (ordinary 750, hard 241). Paired V3-minus-V1 ranges were +26.56–33.16 percentage points recall and -51.13 to -45.21 specificity, with every resample class-defined. The any-finding proxy and source categories limit broad-privacy interpretation.

Updated product audience exactness was 1089/1092/1217 of 1,560; category exactness 885/914/979 and category macro F1 .664/.703/.761. Category-bearing joint was 138/149/237 of

720 and multilabel joint 51/69/125 of 456. Correct unspecified-audience labels decreased by 56/65/21 relative to historical packages; efficient unspecified-audience joint fell from 102 to 98 of 260. Fixed-input repeats matched classifications, actions, evidence and traces.

Post-hoc family resampling gave product binary-accuracy gain ranges 21.64–29.37, 18.28–25.69 and 19.90–29.37 percentage points; expression-group ranges were 23.40–28.08, 19.17–24.81 and 22.44–26.92. These are separate conditional dependence sensitivities, not independent generalization evidence.

APPENDIX D
RELATED-SYSTEM INTERACTION DETAIL

TABLE V
INTERACTION FUNCTIONS REPORTED IN THE REVIEWED VERSIONS. "NOT DESCRIBED" IS LIMITED TO THE CITED VERSION AND REVIEW SCOPE.

System	Prompt and response handling	Warning, acknowledgement and request effect
Casper v1 [9]	Replaces identifiers and restores them in responses.	Topic warning has an acknowledgement control; user can inspect or restore.
HaS v1 [11]	Hides entities and locally restores response entities.	Browser warning acknowledgement and request blocking not described.
ProSan v1 [12]	Selectively rewrites prompt words; evaluates downstream utility proxies.	Browser warning acknowledgement and request blocking not described.
PriVoke	ALLOW/WARN forward original body; returned masked spans are not substituted.	WARN notice is not awaited; BLOCK cancels supported requests.

For ProSan's occupation, race and orientation items, the reviewed v1 further requires WordNet similarity below 0.9; this criterion does not make its item replacement rate a prompt-level privacy metric.

APPENDIX E
SELECTED TRAINING LINEAGE

PH3 fits affine privacy heads over frozen pretrained encoders/poolers using 6,240 TRAIN and 1,560 validation rows, AdamW rate 0.001 and validation-selected checkpoints. EN3 continues each selected parent by jointly fitting encoder/pooler/heads on the same data. AU3 (repository protocol `ba7f3990`) adds 3,120 audience rows to unchanged 6,240-row PH3 rehearsal and uses new 1,560-row validation. EN3/AU3 CPU-four-thread float32 recipes use encoder/pooler rate 2×10^{-5}, head rate 10^{-4}, weight decay 0.01, dropout 0.1 and microbatch 16/effective batch 64. The final incomplete batch uses its actual loss denominator. AU3 warms up 147 updates, allows 5–30 epochs and stops after five validation epochs without strict 10^{-4} loss improvement. Only validation selects trained checkpoints; sealed comparisons select no model, threshold or recipe.

TABLE VI
EXTERNAL REPORTED OUTCOMES ON DIFFERENT TASKS; NO CROSS-STUDY RANKING OR LOCAL REPRODUCTION.

Work	Selected reported outcome	Population / boundary
Casper v1 (2024) [9]	NER recall 98.5%, specificity 86.7%	Two GPT-4-Turbo-generated 2,000-prompt workloads; NER support is 1,000 positive / 1,000 negative.
Casper 2025 [22]	PII accuracy 92.6%	Published abstract: 4,500 synthetic/2,000 real-world prompts; metric support unavailable. Separate from v1.
HaS v1 (2023) [11]	Black-box label-based protection 0.4070	19,703-example train/test corpus, 80/20; 1.7B black-box attacker; one minus whole-text recovery similarity, not entity protection; exact test support unspecified.
ProSan v1 (2024) [12]	Local-star privacy hiding rate 99.8%	MedQA, SAMSum and CodeAlpaca, 1,000 test prompts each; item-replacement proxy, exact item count unspecified.
PromptShield 2025 [23]	Policy accuracy 72.92%	Independent 435-prompt test; full model/class supports unavailable.

Table VII gives selected PIPE3 ancestry. AU3 completed 16/11/9 epochs but selected 11/6/4; discarded work is excluded. Cumulative selected encoder/pooler epochs are 30/20/8 and heads 42/27/12. V1 encoders have zero optimizer-trained epochs and bootstrap heads 220/360/520. These doses are not equivalent to unknown upstream linguistic-pretraining counts.

TABLE VII
SELECTED V3 ANCESTRY. PARAMETERS ARE V1/V3 MILLIONS; PH3 IS HEAD-ONLY, EN3/AU3 JOINTLY TRAIN ENCODER, POOLER AND HEADS.

Tier	Parameters(M)	PH3	EN3	AU3
Efficient	.019/4.785	12	19	11
Balanced	.037/9.596	7	14	6
Quality	.054/17.494	4	4	4

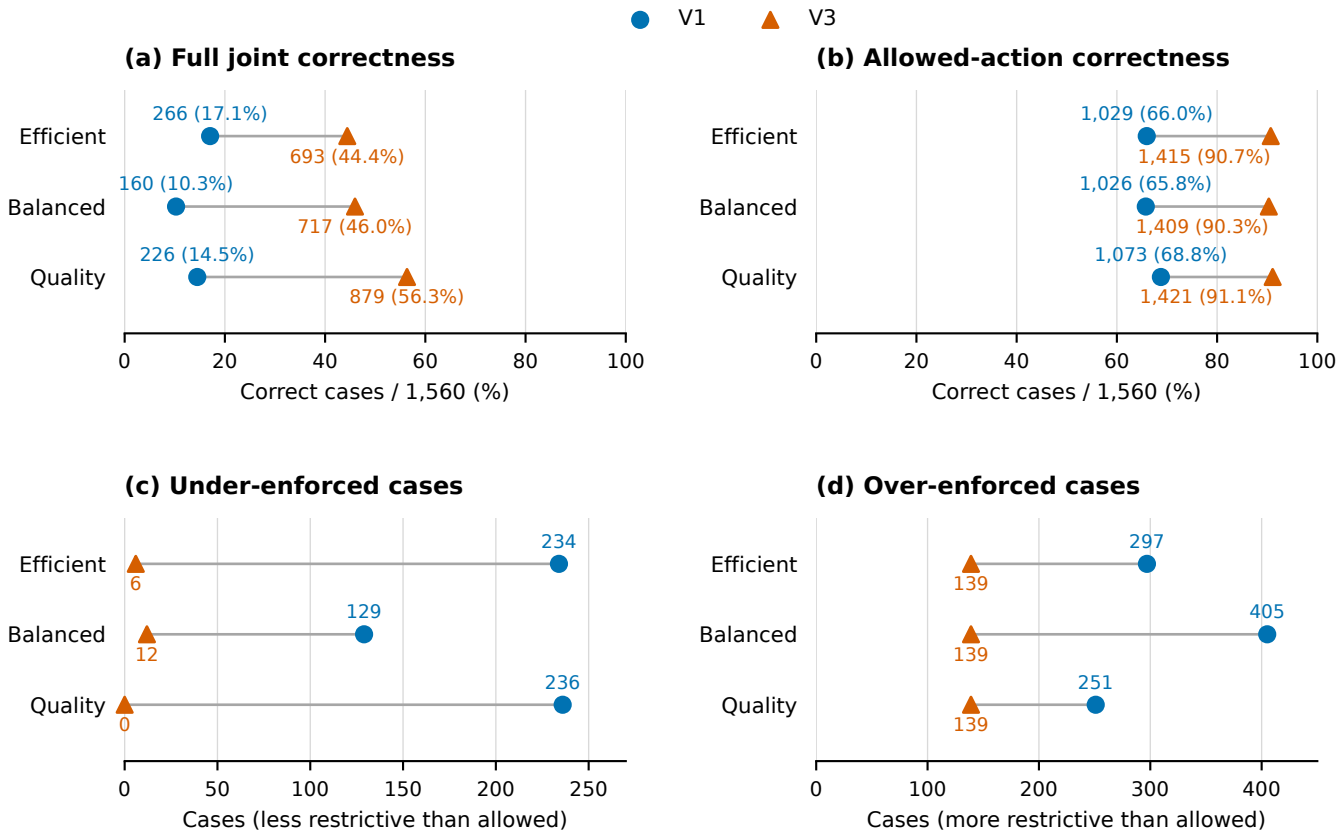


Fig. 4. Detailed product comparison on 1,560 reused provisional cases. V1 denotes historical and V3 updated packages. Under/over-enforced panels count cases on different scales, not ordinal distance; native package differences prevent causal attribution.

Appendix F
Installed-Browser Cost Protocol

The completed cost protocol uses twelve fresh profiles, two per transport/action cell, each with one first decision, five warmups and thirty measured requests selecting regex/NER [2,3]; regex BLOCK can skip NER. All thirty-three correctness/fault probes lie outside resource windows. One container exposes 24 processors without CPU/memory quotas and reuses resident fixture/driver/Go infrastructure. Windows begin before browser launch and end after request/readiness checks, before profile teardown. Thus CPU excludes infrastructure startup before a window, while memory includes resident services/cache. These are instrumented fresh-profile/native-startup measurements, with no unprotected baseline.

Decision latency spans page-observed analysis-request to analysis-result messages, excluding receiver response. Pooled nearest-rank percentiles describe two session clusters; p99 is the observed maximum, without an independent tail interval. The collector requests 100-ms intervals and retains every positive actual gap. CPU-counter differences cover first-to-last observations, with a final read beginning after the last request, boundary checks and stop request. Aggregate cgroup CPU includes descendants; sampled charged memory includes cache/kernel charges rather than RSS/PSS [24]. Totals include Chromium, runtime, Go services, fixture, driver and observer, whose own CPU is retained without subtraction. Known browser/supervisor/detector identities, parent links, running status and listeners are checked at boundaries; continuous process attribution/native ancestry and incremental firewall overhead are unmeasured. Cleanup is required. Nine failed earlier captures remain separate and unpooled.

Appendix G
Descriptive Product Cost

Table VIII reports lower observed V3 first-pass host RPC medians alongside larger process peak RSS. Fixed V3-first ordering, different NumPy/Torch backends, warming and cascade shortcuts preclude controlled speedup or model-only cost claims. Historical service timers and Linux lifetime memory use different scopes.

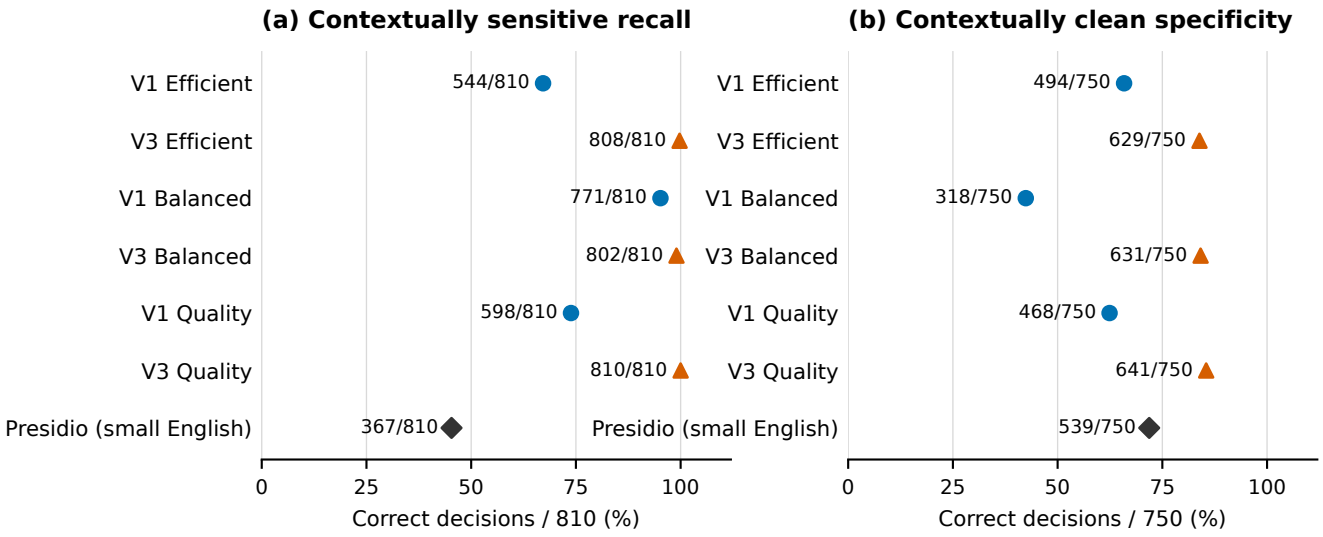


Fig. 5. Same-case binary reference: recall among 810 provisional sensitive and specificity among 750 clean cases. The diamond is small-English Presidio at threshold 0.5. PriVoke-authored reused data and entity/contextual task mismatch preclude an independent privacy ranking.

TABLE VIII
PIPE3 DESCRIPTIVE COST: V1/V3 HOST RPC MEDIAN(P95) MS AND SERVER-PROCESS PEAK RSS MIB. EXCLUDES BROWSER OPERATION.

Tier	Median(p95), V1/V3	RSS, V1/V3
Efficient	55.20(70.08)/43.71(50.12)	389.4/466.8
Balanced	65.87(85.65)/42.73(47.82)	389.8/506.4
Quality	59.58(90.08)/50.02(59.48)	391.8/523.4

APPENDIX H
INSTALLED-BROWSER COST OUTCOMES

The completed protocol yielded twelve fresh-profile windows and 360 measured decisions (Table IX). All 240 ALLOW/WARN measurements executed regex and NER; all 120 BLOCK measurements executed regex and legitimately skipped NER. Nearest-rank p50 values were 1.30 s (range 1.28–9.70) for fresh-profile browser-context creation, 4.37 s (4.21–12.21) for configuration/readiness transition, and 66.4 ms (53.6–75.2) for the first post-readiness decision. These are separate instrumented intervals, not cold-machine or model-only startup.

TABLE IX
INSTRUMENTED PAGE-OBSERVED DECISION LATENCY IN MS FOR THREE FIXED SYNTHETIC CASES, REQUESTED REGEX/NER [2,3]. EACH CELL HAS SIXTY MEASUREMENTS IN TWO FRESH PROFILES; P99 IS THE OBSERVED MAXIMUM. EXCLUDES RECEIVER RESPONSE AND SEMANTIC-LAYER COST.

Transport	Action	p50	p95	p99
fetch	ALLOW	35.0	62.5	73.6
fetch	WARN	31.9	36.1	59.4
fetch	BLOCK	27.9	31.6	33.2
XHR	ALLOW	30.8	35.8	57.7
XHR	WARN	31.3	34.6	59.2
XHR	BLOCK	28.5	32.2	34.1

Across the twelve observed windows (126.40 s summed spans), aggregate container CPU was 193.29 s, including 0.69 s sampled observer CPU. Maximum sampled cgroup charged memory was 1,267.91 MiB, including resident services/cache. The 1,236 samples produced 1,224 within-window gaps: median/p95/maximum 101.21/102.38/622.84 ms, with one over 500 ms. Cumulative CPU and sampled charged-memory maxima differ from wall time and process RSS/PSS; gaps can hide memory spikes. Final-read coverage and process/listener cleanup passed, with natural container exit zero. Whole-container/instrumented scope does not isolate client overhead or support speed comparisons with Table VIII.

APPENDIX I
TELEMETRY AND LOCAL RECONSTRUCTION DETAILS

A fingerprinted Windows CPython 3.11.9 audit enumerated exact rational finite kernels for the actual five domains, $K = 3, 4, 11, 2, 6$. Outward 100-digit logarithm enclosures place maximum event loss above nominal $\varepsilon = 0.5, 1, 2$ by approximately 1.294×10^{-15}, 1.503×10^{-15} and 1.226×10^{-15},

respectively. Thus the exact nominal bound fails in these settings, despite the tiny excess. Eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same entropy assumptions. Other configurations and deployed builds remain uncertified.

Across 180 fixed-truth simulation cells and 10,000 repetitions per cell, inversion and clipping produced substantial conditional error. For balanced integer category counts with $K = 11$, event $\varepsilon = 1$ and $n = 80/8000$ contributing reports, first-category true counts were $7/727$, clipped bias $+22.84/+237.50$ and RMSE $42.32/1028.47$ counts. Clipping occurred in $85.62/28.32\%$ of trials; independently estimated marginals failed to sum to n in $99.99/99.78\%$. Larger n pools installation-days, each capped at eight default reservations. These are errors around realized contributor counts, excluding population sampling; simulator probabilities are binary64 approximations of the exact finite kernel. Numerical moment/collector checks support implementation consistency, not a deployment-usefulness threshold.

Real isolated ledger checks accepted eight reservations and denied the ninth, preserved the cap across process reopen, reset next UTC day, failed closed on rollback/corruption and accepted sixteen of forty concurrent epsilon-0.5 requests. Build, queue and delivery failures consumed reservations. A constructed 32-event sequence with 25% BLOCK yielded 100% versus 0% BLOCK among first-eight contributors when event order reversed. Randomization therefore does not remove contribution-selection bias; no real deployment monitoring accuracy was measured.

A. Local Source/Model Reconstruction

A separately sealed 195-pin local capsule copies the 158 runtime files actually listed in the later FN14 freeze, additional helpers, six frozen V1/AU3 model bundles and prior invented probe receipts. This inventory is distinct from the 154-file historical clean-checkout comparison. Six fresh isolated-source host servers replayed the same four invented sole-semantic requests per package: all 24 classifications/actions and four package identity fields matched, with zero errors. Independent post-run checks verified all pins, import origins and process cessation. This closes a bounded local reconstruction gap using the maintained Python environment; it is not a public Git-only installation, new dependency environment, accuracy test or protected-final rerun.

B. Implemented Finite Kernel

Let $\widetilde{p}_i$ be the binary64 truth threshold. In the audited CPython sampler, fresh uniform 53-bit J selects truth when $J/2^{53} < \widetilde{p}_i$. Hence

$$P_i = \frac{\lceil 2^{53}\widetilde{p}_i \rceil}{2^{53}}, \quad Q_i = \frac{1 - P_i}{K_i - 1}, \\ \varepsilon_{\text{finite}} = \sum_i \log \max\{P_i/Q_i, Q_i/P_i\}. \tag{2}$$

Alternative selection uses rejection sampling. The calculation assumes independent uniform OS bits and the fingerprinted host arithmetic; it does not certify deployed entropy or other builds.

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